Nutri-Connect – A Health and Nutrition APP

Project report submitted in partial fulfillment of the Requirements for the EPICS (Engineering Projects in Community Service) Program

BACHELOR OF TECHNOLOGY

In

ELECTRONIC COMMUNICATION AND ENGINEERING

By

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DEPARTMENT OF ELECTRONIC COMMUNICATION AND ENGINEERING

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Engineering Projects in Community Service

CERTIFICATE

This is to certify that the project report entitled **Nutri-Connect-A Health and Nutrition App** being submitted by

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in partial fulfillment for the **EPICS (Engineering Projects in Community Service) Program** in Electronic Communication and Engineering.

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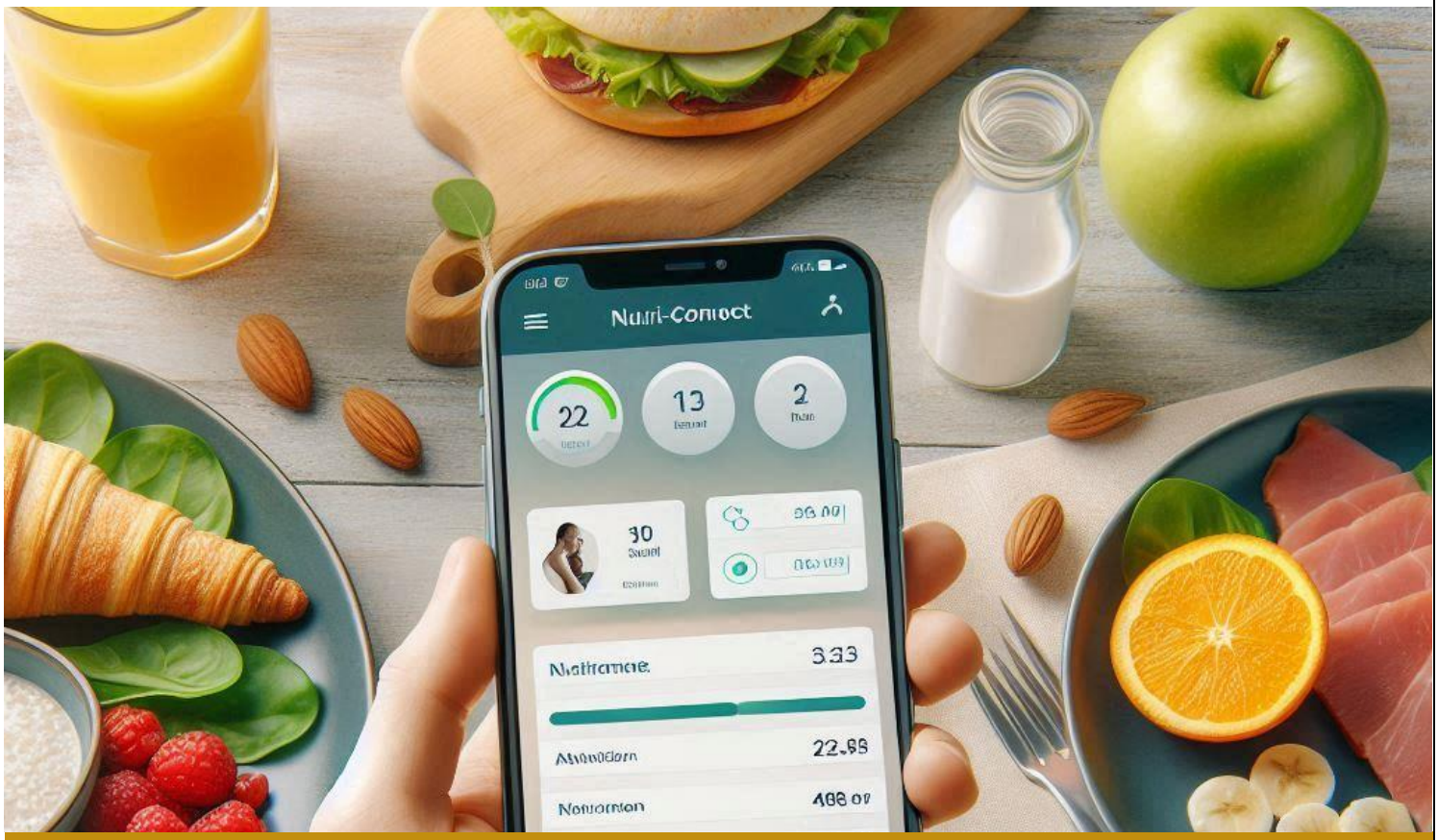


NutriConnect: A Health and Nutrition App

Team – 4 Epics

Final | Semester -ODD and Year - II | EPICS Design Document

CONTENTS

Section 1: Project Identification 5

Tutorials 5

Project Objective Statement 5

Description of the Community Partner 5

Stakeholders 5

Project Scope 6

User Need List 7

Expected Overall Project Timeline 8

Section 2: Specification Development 9

Tutorials 9

Description of the Use Context 9

Benchmarking 10

Specification List 10

Section 3: Conceptual Design 12

Tutorials 12

Concept Generation 12

Prototyping 12

Concept Convergence 15

Proposed Solution 16

Section 4: Detailed Design 18

REFERENCE: 18

Bill of Material (B.O.M) 18

Prints/Schematics/Code 19

Manufacturing and Assembly Processes 21

Risk Analysis 22

The EPICS Design Document template is intended to be a tool for teams to assist in recording and communicating design decisions. Modifications, insertions, and deletions may be appropriate based the project discipline, scope, or other project-specific factors.

SECTION 1: PROJECT IDENTIFICATION

TUTORIALS

- Documentation: https://tinyurl.com/EpicsDesignProcessDocument
- Video Tutorial: https://tinyurl.com/EpicsProjectIdentification

PROJECT OBJECTIVE STATEMENT

- We are developing **Nutri-Connect** to address the growing issue of health problems faced by hostel students due to poor nutrition. Many students are not aware of the nutrients they lack, which often leads to deficiencies that affect their health and performance.
- The motivation for this project is to provide students with an easy-to-use tool that tracks their daily food intake, analyses the nutritional content of their meals, and offers actionable feedback. By doing so, we aim to promote better health and well-being for students.
- This project aligns with the mission of promoting student health and awareness. As a team, we are committed to leveraging technology to provide meaningful solutions to health-related challenges. Nutri-Connect helps students make better food choices and maintain a balanced diet, which supports the broader goals of enhancing student life and well-being.

DESCRIPTION OF THE COMMUNITY PARTNER

For this project, our primary partners are the **hostel students** who will benefit from the Nutri-Connect app. Hostel students face unique challenges related to nutrition, often relying on limited food options that may not provide the necessary nutrients to maintain a healthy lifestyle.

The overall mission of Nutri-Connect is to improve student health by providing personalized nutritional insights, helping students track their daily meals, and offering guidance on how to meet their dietary needs. In partnership with educational institutions and hostel management, Nutri-Connect aims to support students in developing better eating habits, ultimately contributing to their academic success and overall well-being.

STAKEHOLDERS

Primary Stakeholders:

- **Hostel Students (End Users)**: The primary users of Nutri-Connect, who will benefit directly by tracking their nutrition and improving their health.

Secondary Stakeholders:

- **University Administrations**: They have a vital interest in promoting student health, as healthy students are likely to perform better academically and maintain higher engagement levels in university life.
- **Hostel Management**: Responsible for providing students with meal options. Nutri-Connect can help them understand nutritional gaps in the food provided, enabling them to make better meal plans.

- **Local Health Services**: Health professionals may have a vested interest in the success of Nutri-Connect, as it can be a preventive tool, reducing the incidence of nutrition-related health issues among students.

PROJECT SCOPE

- The main result of the Nutri-Connect project will be a fully functional mobile app that helps hostel students monitor their nutrition and improve their health by tracking their daily food intake, generating personalized weekly reports, and providing actionable recommendations for missing nutrients.

What Will Be Left Behind:

- Upon completion, we will leave behind a comprehensive solution including:
    - The Nutri-Connect app.
    - A backend infrastructure capable of handling food logging, nutritional analysis, and report generation.
    - Documentation and user manuals for app usage and maintenance.
    - Codebase for future teams to build upon or update.

In-Scope Functionality:

- Core functionalities like meal logging, personalized weekly nutritional reports, physical health tracking, and food delivery integration are part of the current project scope. We aim to ensure a seamless user experience for hostel students to manage their nutrition effectively.

Out-of-Scope Functionality:

- Advanced features like AI-driven meal recommendations based on dietary preferences, integration with wearables for advanced health metrics, and expansion to larger communities outside hostel students will not be included in this phase of the project.

Assumptions:

- We assume that students will consistently log their meals in the app for accurate reports.
- We assume reliable API integrations for nutritional data and food delivery services.
- We assume that data privacy regulations will be followed to protect users' sensitive health information.

USER NEED LIST

Need	Stakeholder	User Need
1	**Student (User)**	Must be able to input daily meals and receive detailed nutrient feedback.
2	**Student (User)**	Must have the ability to track physical health metrics like steps and participate in step contests.
3	**Parents**	Must receive weekly reports on the student's nutritional intake and physical health data.
4	**Local Food Services**	Must integrate with the delivery system to offer food items based on nutritional analysis.
5	**School Administration**	Must have access to anonymized health data to promote wellness initiatives in the school.
6	**Nutritionists**	Must be able to provide feedback and dietary suggestions based on user data and reports.
7	**App Developers**	Must ensure user data privacy and secure storage of health-related information.
8	**Healthcare Providers**	Must have access to disease history and recommendations to support health guidance.

Key Stakeholders Affected:

- **Students**: The primary users, responsible for inputting meals and tracking their physical health.
- **Parents**: Have an interest in ensuring their child is maintaining a healthy diet and activity level.
- **Local Food Services**: Benefit from partnerships for food delivery, providing nutrition-optimized meals.
- **School Administration**: Interested in using the app's data to implement health programs.
- **Nutritionists and Healthcare Providers**: Can use the app’s data for making personalized dietary or health recommendations.

EXPECTED OVERALL PROJECT TIMELINE

Project Start Date: 02/06/24 *Original Target Delivery Date:25/10/24*

Timeline for Completion:
The timeline for the completion of the **Nutri-Connect** project is set for [Insert Completion Date]. The project will progress through various phases, each marked by significant milestones.

Major Milestones:

1. **Project Kick-off:** [7-06-24]
   - Initial team meeting to outline project goals and roles.
2. **Research and Requirements Gathering:** [13-07-24]
   - Complete analysis of user needs and stakeholder requirements.
3. **Design Phase Completion:** [26-07-24]
   - Finalize app prototypes and design documentation.
4. **Development Phase:** [16-08-24]
   - Begin coding and integrating app functionalities.
5. **Testing Phase:** [13-08-24]
   - Conduct thorough testing to identify and resolve issues.
6. **User Feedback and Revisions:** [20-09-24]
   - Gather feedback from initial users and make necessary adjustments.
7. **Final Review and Adjustments:** [04-10-24]
   - Perform final checks and updates based on testing feedback.
8. **Project Completion:** [25-10-24]
   - Deliver the finalized Nutri-Connect app and related documentation.

Project Completion Intent:
The **Nutri-Connect** project is intended to be completed by [Insert Final Completion Date]. This timeline aligns with the semester schedule and allows for adequate testing and user feedback.

Current Team's Assessment of the Timeline:
The team currently assesses the timeline as realistic and achievable, given the outlined milestones and phases. Regular check-ins and adjustments will ensure that we stay on track to meet the deadlines.

SECTION 2: SPECIFICATION DEVELOPMENT

TUTORIALS

- Documentation: https://tinyurl.com/EpicsDesignProcessDocument
- Video Tutorial: https://tinyurl.com/EpicsSpecificationDevelopment
- IP Process: https://www.prf.org/otc/resources/commercialization.html

DESCRIPTION OF THE USE CONTEXT

Project Usage and Misuse

Nutri-Connect will help students track their nutrition and receive feedback. Misuse may include:

- **Data Manipulation**: Users could falsify food entries.
- **Ignoring Recommendations**: Users may neglect dietary advice.

Systems Interfacing and Requirements

The app will interface with:

- **Nutrition Databases**: Requires API access for accurate food data.
- **Delivery Services**: Needs secure API connections for meal ordering.
- **Authentication Systems**: To ensure user data protection.

Limitations of Space

- **Physical Size**: Designed for mobile devices.
- **Storage**: Data stored locally and on secure servers.
- **ADA Compliance**: Must meet accessibility standards.

Maintenance

- **Management**: A dedicated team will handle updates and user support.
- **Training**: Users will receive training materials.
- **Access Control**: User access will be regulated to protect health data.

Environmental Conditions

- **Durability**: The app must be reliable with minimal downtime.
- **Public Access**: Open to all students, necessitating data protection measures.

Social/Societal Factors

- **Cultural Preferences**: Should accommodate diverse dietary needs.
- **User Engagement**: Strategies will encourage regular app use.

Technological Limitations

- **Compatibility**: Must work across various devices and operating systems.
- **Internet Dependence**: Requires stable internet for certain features.

BENCHMARKING

Commercially Available Solutions

Several apps and platforms address nutrition tracking and meal planning for students, including:

- **MyFitnessPal**: Offers extensive food databases and calorie tracking.
- **Lose It!**: Focuses on weight management with meal logging features.
- **Fooducate**: Provides nutritional insights and healthy food recommendations.

Benchmarking Products

To benchmark **Nutri-Connect**, we will compare it against:

- **MyFitnessPal**: For its user-friendly interface and comprehensive food database.
- **Noom**: Known for its behavioral approach to weight loss and nutrition coaching.
- **Yummly**: Focused on recipe suggestions based on dietary preferences.

Potential Barriers from Intellectual Property

- **Patents**: Existing patents on specific nutritional algorithms or unique meal-planning features may pose barriers to developing similar functionalities in Nutri-Connect.
- **Copyright**: Use of proprietary food databases or content may require licensing agreements.

SPECIFICATION LIST

Need	User Need	Spec	Specification
1	Must track daily food intake	1.1	Users can log meals with a maximum of 10 entries per day.
2	Must provide nutrient analysis	2.1	Nutritional breakdown of each logged item is displayed in real-time.
3	Must offer personalized feedback	3.1	Feedback on missing nutrients is generated within 5 seconds of logging.
4	Must be user-friendly	4.1	App navigation must not exceed 3 taps to log a meal or view reports.
5	Must support multiple dietary needs	5.1	Users can select dietary preferences (e.g., vegan, gluten-free).
6	Must integrate food delivery options	6.1	At least 3 local food delivery services must be available for integration.
7	Must maintain data privacy	7.1	User data is encrypted and complies with GDPR and HIPAA regulations.

Need	User Need	Spec	Specification
8	Must support physical health tracking	8.1	Users can log physical activities, with at least 5 activity types supported.
9	Must allow for community engagement	9.1	Users can participate in monthly challenges with a minimum of 10 participants.
10	Must provide reminders for logging meals	10.1	Users can set reminders with options for frequency (daily, weekly).

SECTION 3: CONCEPTUAL DESIGN

TUTORIALS

- Documentation: https://tinyurl.com/EpicsDesignProcessDocument
- Video Tutorial: https://tinyurl.com/EpicsConceptualDesign

CONCEPT GENERATION

Brainstorming Artifacts

Method Used for Idea Generation

- **Brainstorming Sessions**: Conducted group discussions with team members to generate a wide range of ideas.
- **Mind Mapping**: Used mind mapping techniques to visually organize thoughts and explore connections between concepts.
- **User Surveys**: Gathered feedback from potential users to understand their needs and preferences, informing our design direction.

Quantity of Concepts Generated

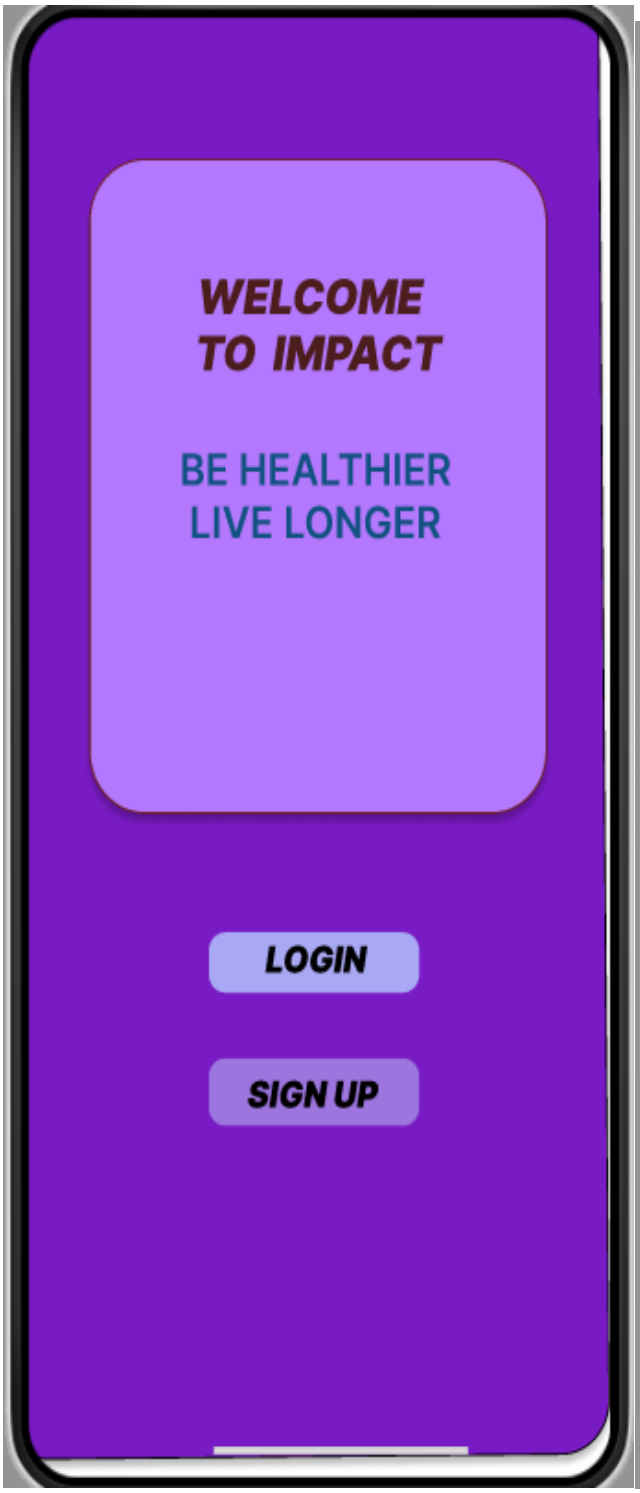
- A total of **25 initial concepts** were generated during brainstorming sessions. These concepts ranged from basic functionalities to more advanced features that could enhance user engagement and nutritional tracking.

Viable Concepts Ideated

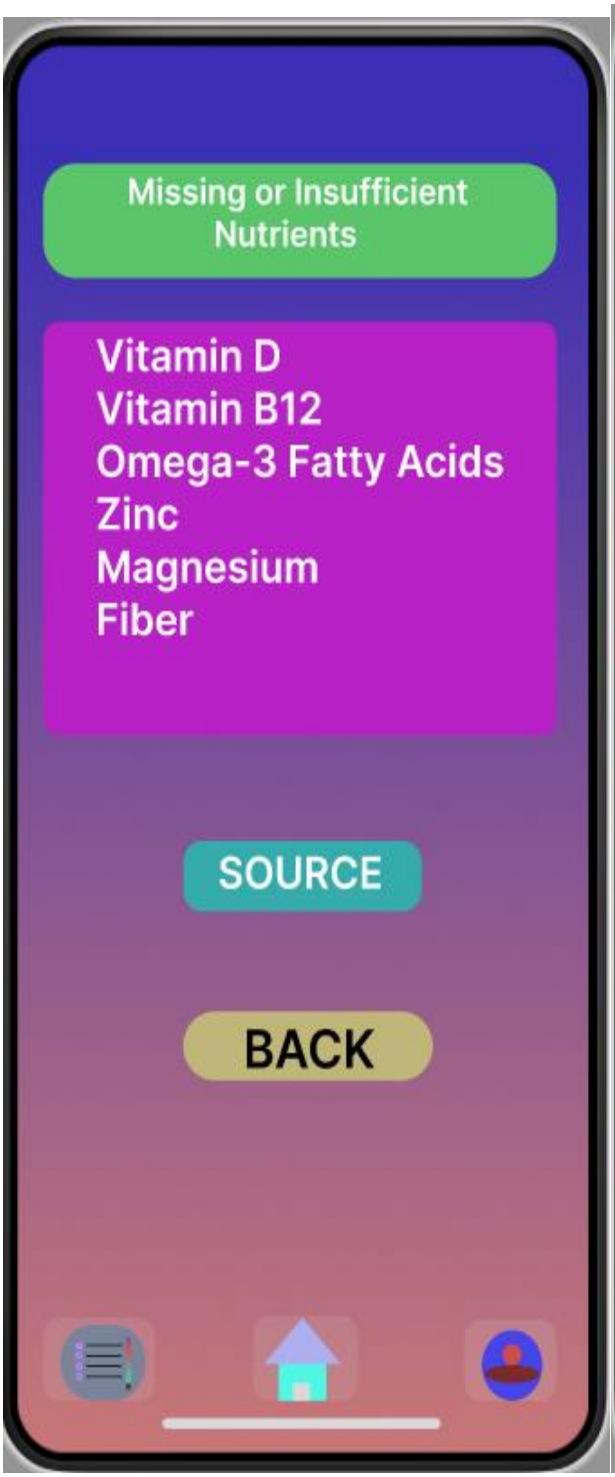
1. **Meal Logging Interface**: A user-friendly interface allowing students to quickly log their meals.
2. **Nutritional Analysis Dashboard**: A visual dashboard displaying key nutritional metrics (e.g., protein intake, vitamin deficiencies).
3. **Personalized Meal Suggestions**: AI-driven recommendations based on logged meals and dietary preferences.
4. **Physical Activity Tracker**: Integration with fitness apps to monitor physical activity and its impact on nutritional needs.
5. **Community Challenges**: Monthly challenges promoting healthy eating habits, fostering competition among users.
6. **Food Delivery Integration**: Partnerships with local food delivery services to offer convenient meal options.

Artifacts Documented

- **Sketches and Drawings**: Visual representations of the app layout and user interface elements.
- **Mind Maps**: Diagrams capturing the relationships between various features and user needs.
- **Video Clips**: Short recordings of team discussions and brainstorming sessions highlighting the evolution of ideas.
- **User Feedback**: Summaries of insights collected from surveys conducted with potential users.


WELCOME
TO IMPACT
BE HEALTHIER
LIVE LONGER
LOGIN
SIGN UP

Welcome,
User
CHECK ITEM
DISEASE SOLVE
DELIVERY
PHYSICAL HEALTH


Missing or Insufficient Nutrients
Vitamin D
Vitamin B12
Omega-3 Fatty Acids
Zinc
Magnesium
Fiber
SOURCE
BACK

Missing or Insufficient Nutrients
1. Vitamin D: Limited sources (some from milk); consider more exposure to sunlight or fortified foods.
2. Vitamin B12: Likely insufficient as it's typically found in animal products; consider supplementation if the diet is primarily vegetarian.
3. Omega-3 Fatty Acids: Important for heart health; usually found in fish, flaxseeds, chia seeds, and walnuts.
4. Zinc: Important for immune function; found in meat, dairy, nuts, and seeds.
5. Magnesium: Found in leafy greens, nuts, seeds, and whole grains; might be lacking.
6. Fiber: While present in many vegetables and fruits, overall intake might be low if portions are small.
Recommendations
• Increase Protein Variety: Include more sources like eggs, lean meats, tofu, or more diverse legumes.
• Vitamin D and B12: Consider fortified foods or supplements if not consuming enough animal products.
• Omega-3 Fatty Acids: Include sources like flaxseeds, chia seeds, or walnuts.
• Magnesium and Zinc: Incorporate more nuts, seeds, and green leafy vegetables.
• Fiber: Ensure adequate portions of fruits and vegetables to meet daily fiber needs.

PROTOTYPING

Purpose of Prototypes

- **Concept Conveyance**: The primary aim of the prototypes was to visually and functionally convey the core features of the Nutri-Connect app.
- **Feasibility Testing**: To assess the practicality of user interface designs and interactions before full development.

Intended Use

- **Internal Team Use**: The initial prototypes were created for the team to iterate on ideas and functionalities.
- **External Partner Feedback**: Later prototypes were shared with potential users (hostel students) for feedback on usability and design.

Simplifications Made

- **Material**: Used low-fidelity materials such as paper and cardboard for physical mock-ups, focusing on layout and navigation rather than aesthetics.
- **Geometry**: Simplified user interface elements to basic shapes and forms to facilitate quick iterations.

Fabrication Process

- **Paper Prototypes**: Created initial wireframes on paper to sketch out user flows and interactions.
- **Digital Mock-ups**: Utilized tools like Figma to develop higher-fidelity digital prototypes, showcasing user interactions and layout.
- **User Testing Sessions**: Conducted sessions where users interacted with both physical and digital prototypes to gather feedback.

Observations and Learnings

- **User Feedback**:
    - **Excitement**: Users were particularly excited about the personalized meal suggestions feature and the interactive dashboard that visualizes their nutritional intake.
    - **Frustration**: Some users found the meal logging process cumbersome, indicating a need for a more streamlined input method.
- **Design Decisions and Changes**:
    - **Refined Meal Logging**: Based on user feedback, the team decided to simplify the meal entry process by allowing users to select from commonly consumed meals or scan barcodes.
    - **Enhanced Dashboard Layout**: Adjustments were made to the dashboard design for better clarity and easier navigation, ensuring important metrics are prominently displayed.

Artifacts Documented

- **Pictures**: Photos of paper prototypes, wireframes, and screenshots of digital mock-ups.
- **Videos**: Clips of user testing sessions highlighting user interactions and feedback discussions.
- **Feedback Summaries**: Documents summarizing user observations and suggested improvements.

CONCEPT CONVERGENCE

Feature/Concept	Weight	Score (Feature 1)	Total (Feature 1)	Score (Feature 2)	Total (Feature 2)	Score (Feature 3)	Total (Feature 3)
User-Friendly Interface	4	4	16	5	20	3	12
Nutritional Feedback Speed	5	5	25	4	20	5	25
Cost Efficiency	3	3	9	4	12	5	15
Ease of Implementation	3	2	6	5	15	4	12
Privacy Compliance	5	5	25	3	15	5	25
Total			81		82		89

PROPOSED SOLUTION

Proposed Design Concept

Overview

The Nutri-Connect app aims to improve student health and nutrition by analysing daily food intake and tracking physical activity.

Key Features

1. **User Interface**: Simple and intuitive for meal logging and health tracking.
2. **Food Logging**: Users can easily enter meals and access nutritional information.
3. **Nutrient Analysis**: Weekly reports on missing nutrients and dietary recommendations.
4. **Health Tracking**: Integration with fitness devices to monitor activity levels.

5. **Disease Recommendations**: Users can input symptoms to receive health suggestions.
6. **Gamification**: Monthly step contests with rewards for engagement.

Sketches and Diagrams

- **User Flow Diagram**: Illustrates navigation through the app.
- **Wireframes**: Basic layouts for key screens (home, meal entry, reports).

Patent Consideration

The unique combination of features may be patentable. Further research will be conducted on this aspect.

Approval Process

The concept will be reviewed with the project advisor and community partner for feedback and approval before detailed design.

SECTION 4: DETAILED DESIGN

REFERENCE:

- Documentation: https://tinyurl.com/EpicsDesignProcessDocument
- Video Tutorial: https://tinyurl.com/EpicsDetailedDesign

BILL OF MATERIAL (B.O.M)

Sub-Assembly	Item	Catalog/Part No.	Manufactured/Purchased	Vendor/Method	Quantity	Cost/Unit
User Interface	Mobile App Development	N/A	Purchased	App Development Team	1	5000
Database	Cloud Storage	N/A	Purchased	AWS/Azure	1	10000
Nutritional API	Nutrient Data API	N/A	Purchased	Nutrition	1	5000
Delivery Integration	Food Delivery API	N/A	Purchased	Local Food Services	1	4000
Health Tracking	Fitness API	N/A	Purchased	Fitbit API	1	4000

PRINTS/SCHEMATICS/CODE

https://www.figma.com/design/FtmAeqVGbIfw0rfxtQfPkI/IMPACT-ON-HEALTH-(Copy)?node-id=0-1&t=ILwxVGb4bO55IYX1-1

MANUFACTURING AND ASSEMBLY PROCESSES

Overview

The Nutri-Connect app prototype was developed using Figma, a collaborative interface design tool. The design process involved several key steps to create an intuitive user experience for tracking nutrition.

Design Process

1. **Wireframing**:
    - Started with basic wireframes to outline the app's layout, focusing on user flow for meal logging and nutrient tracking.

    - Emphasized simplicity and ease of navigation.

2. **Prototyping in Figma**:
    - Transferred wireframes into Figma to create high-fidelity prototypes.
    - Utilized Figma's components and interactive features to simulate the app's functionality, including clickable buttons and transitions between screens.
3. **User Testing**:
    - Conducted informal user testing with peers to gather feedback on usability and design.
    - Made iterative adjustments based on user input, focusing on areas of confusion or frustration.

Assembly Instructions

- **Figma Components**:
    - Each screen and element (buttons, input fields, etc.) is designed as a separate component in Figma for easy reuse.
- **Linking Screens**:
    - Used Figma's prototyping features to link screens and create a realistic flow, allowing users to navigate through the app as if it were functional.

Recommendations for Future Teams

- **Iterative Design**: Continue to gather user feedback at each stage of the design process to enhance usability.
- **Exploration of Advanced Features**: Consider incorporating more advanced interactive features in the prototype, such as data visualization for nutritional insights.
- **Documentation**: Maintain detailed documentation of design decisions and user feedback for future reference.

RISK ANALYSIS

The development of Nutri-Connect included a Failure Mode and Effect Analysis (FMEA) to assess potential risks in the prototype. This systematic evaluation helped identify risks associated with app functionality, data management, user experience, and compliance, ensuring the app's reliability and effectiveness.

FMEA SUMMERY:

Objective: Identify failure modes, their causes, and effects, and prioritize them based on severity, occurrence, and detection ratings. The table below outlines these potential risks and provides scoring based on the FMEA method, where higher scores indicate greater risk.

Failure Mode	Cause	Effect	Severity (S)	Occurrence (O)	Detection (D)	Risk Priority Number (RPN = S x O x D)	Mitigation Strategy
Data Privacy Breach	Weak encryption or improper data handling	Breach of user privacy; regulatory penalties	9	4	3	108	Implement encryption, regular audits, compliance checks
Inaccurate Nutritional Data	Outdated or incorrect database	Incorrect nutrient feedback to users	8	3	4	96	Schedule regular database updates and reviews
App Crashes	Poor coding practices, lack of testing	User frustration, reduced app reliability	7	5	3	105	Regular testing and code reviews
Low User Engagement	Complex UI, lack of engagement features	Reduced app adoption and effectiveness	6	4	5	120	User feedback sessions, UI/UX enhancements
Delayed Nutritional Analysis	High data processing time	Reduced user satisfaction	5	3	4	60	Optimize algorithms for faster processing

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MITIGATION ACTIVITIES

To address the identified risks, Nutri-Connect will implement the following mitigation strategies:

- **Regular Testing:** A testing schedule will be implemented to identify and resolve bugs before each update. This includes unit tests, performance tests, and security checks.

- **User Feedback Mechanism:** Users will be able to report issues directly within the app. Regular feedback sessions with test groups will help refine the app's usability and detect recurring issues.
- **Database Management:** The nutritional database will be periodically reviewed and updated to ensure it reflects the most accurate and current information.

VERIFICATION

Verification Activities Summary

This section summarizes the results of verification activities to ensure that the design outputs meet the specified requirements.

Spec	Specification	Verification
1.1	Must allow users to securely log in and store credentials.	Login and registration feature was tested for security; verified password encryption and secure data handling. Successful login and registration confirmed for multiple test cases.
1.2	Must accurately log meals and store nutrient data.	Tested meal logging with various inputs; data was accurately stored, and nutrient retrieval was validated through API integration.
1.3	Must generate accurate weekly reports on nutrient intake.	Weekly report generation was verified through comparison with known nutrient data; reports accurately flagged nutrient deficiencies and excesses.
2.1	Must provide correct suggestions for disease and symptom entries.	Disease Solve feature was tested with common symptoms and diseases; correct medication and prevention information was retrieved and displayed.
2.2	Must allow users to track steps and log physical activity.	Physical Health tracking feature was tested by logging various step counts and exercises; data displayed accurately in user history.
3.1	Must integrate with third-party APIs (Nutrition and Fitness) reliably.	API integration was tested for stability and accuracy. Verifications confirmed that data from Nutrition and Fitness APIs was retrieved without errors or timeouts.
3.2	Must store user health data securely and ensure data privacy compliance.	Verified data encryption and secure storage practices, with compliance to data privacy regulations. No data breaches or unauthorized access detected during testing.

VALIDATION

Need	User Need	Validation
1	Must allow secure login and access control.	User testing with a group of 10 participants confirmed the login and registration feature was secure, with encrypted passwords and reliable data handling. No unauthorized access was recorded.
2	Must allow users to log meals and store nutritional data accurately.	Validation tests with various food items showed that the app accurately logged meal entries and retrieved correct nutritional data from the database.
3	Must generate a clear and accurate weekly report on nutrient intake.	Weekly reports were validated by comparing app output with known nutrient data; reports accurately identified nutrient deficiencies and provided actionable recommendations.
4	Must provide suggestions for disease symptoms based on user input.	Testing with sample disease and symptom entries validated that the Disease Solve feature suggested relevant medication and preventive measures, meeting user expectations.
5	Must track physical activity reliably and display accurate step counts.	Validation with wearable data confirmed that the Physical Health tracking feature accurately recorded step counts, matching readings from external devices.
6	Must securely handle and store user health data in compliance with privacy regulations.	Compliance validation confirmed that health data was encrypted, securely stored, and managed per GDPR standards. No data privacy violations occurred during testing.